

FPGA-Based ECG Arrhythmia Classification Using Dual-Branch DSCResNet with Inter-Patient Evaluation

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Abstract—This paper presents a compact Two-Branch Depth-wise Separable Convolutional ResNet (DSCResNet) for five-class AAMI ECG arrhythmia classification on the Xilinx Zynq-7020 FPGA. The architecture jointly models waveform morphology through a CNN branch and eight-feature RR interval timing through an RR-MLP branch, fusing both into a 160-dimensional representation for final classification. Unlike all six compared works, this paper strictly enforces the AAMI EC57-compliant inter-patient evaluation protocol, partitioning the MIT-BIH Arrhythmia Database by patient identity with no beat-level overlap between partitions, ensuring all reported metrics represent genuine generalisation to unseen patients. The model achieves 83.11% an overall accuracy and a macro F1-score of 0.521, with a Ventricular ectopic recall of 79.3% and a Supraventricular ectopic recall of 43.2%. The Supraventricular recall attributed entirely to the RR-MLP branch, with ablation experiments confirming that waveform morphology alone is insufficient for S-class detection. At only 0.43 MB, the model is the most compact in the comparison. A 4th-order Butterworth bandpass filter implemented in Verilog as four cascaded Q16.16 Second-Order Sections shares exact coefficients with the MATLAB training pipeline, consuming 2,326 LUTs (4.37%), 516 flip-flops (0.48%), zero DSP48 blocks, and 29 mW of logic power, operating at 116.6 MHz with all timing constraints satisfied. Future work will extend the implementation to include the full neural network inference accelerator on the PYNQ-Z2 platform.

Index Terms—ECG arrhythmia classification, FPGA, Butterworth filter, AAMI EC57, Edge deployment.

I. INTRODUCTION

One of the leading fatal disease globally is Cardiovascular disease, which causes approximately 17.9 million deaths annually. The electrocardiogram (ECG) remains the primary non-invasive tool for detecting cardiac arrhythmias, and automated beat classification is essential for continuous monitoring where clinician review of every beat is impractical. Recent FPGA-accelerated systems have demonstrated accuracy figures reaching 96 – 99% on this task [1] - [7], however deep learning has shown strong performance but two fundamental problems limit the clinical utility of these published results [8]. First, the majority of works evaluate their models under random beat-level splits, where the data of same patient's beat appear in both training and test sets. This constitutes data leakage and the resulting accuracy figures overestimate genuine generalisation to new patients [9], [10]. The AAMI EC57 standard explicitly prohibits this approach and mandates inter-patient partitioning [11]. Second, accurate real-time inference on hardware requires that the signal conditioning applied during inference exactly matches the preprocessing used during training. Any mismatch shifts the input distribution and degrades accuracy [12], [13], [14]. This paper addresses both limitations. We propose the Two-Branch DSCResNet, a novel architecture that jointly models waveform morphology and eight-feature RR interval timing for five-class AAMI classification, evaluated exclusively under the inter-patient protocol [15], [16]. Sepa-

rately, we implement the training-time Butterworth bandpass filter as a verified Verilog hardware module characterised across four synthesis metrics on the Xilinx Zynq-7020 device.

- (i) The first dual-branch architecture jointly modelling waveform morphology and multi-feature RR timing for five-class AAMI classification..
- (ii) The only work in the comparison enforcing AAMI EC57-compliant inter-patient evaluation.
- (iii) The smallest model footprint (0.43 MB) with a verified hardware preprocessing filter, establishing training-inference consistency for edge deployment.

II. REPORTED DESIGNS

Recent FPGA-accelerated ECG classifiers have demonstrated high accuracy but share common methodological limitations. Cao et al. [1] proposed a quantization-aware training framework with a cosine-approximated CWT module and a six-layer Inception-ResNeXt network, achieving 99.5% on MIT-BIH with 200 mW dynamic power. The system uses no RR interval features and evaluates under a random beat split. Lu et al. [2] implemented a CNN-based classifier achieving 98.9% using bandpass-filtered signals with two RR features, again under random split conditions. Akshayraj et al. [3] presented an energy-efficient CNN for wearable devices achieving 99.1%, also with two RR features and random split evaluation. Aruna et al. [4] accelerated a deep CNN on FPGA reporting 96.5% under identical random split conditions. Liu et al. [5] proposed a lightweight 1D-CNN with depthwise separable convolutions achieving 99.59% in software and demonstrated FPGA deployment, using no RR features and random split evaluation. Sekhar et al. [6] based on a hybrid CNN-LSTM model using three RR features and achieved 99.0% on MIT-BIH, again without inter-patient separation. The inter-patient evaluation protocol was established by Ali et al. [9], who demonstrated that heartbeat classification systems must be validated on patient recordings unseen during training to provide clinically meaningful performance estimates [17], [18]. None of the six compared works follow this protocol; the proposed system is the first in this comparison to do so.

III. DATASET AND PREPROCESSING

A. Dataset

The MIT-BIH Arrhythmia Database which contains 48 two-lead ECG recordings and sampled at a rate of 360 Hz from 47 subjects with beat-level annotations. Following the AAMI EC57 standard, beats are grouped into five classes: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), Fusion (F), and Unknown (U). The dataset is partitioned strictly by patient identity, 34 patients for training, 7 for validation, and 8 for testing. Test patients are drawn exclusively from the 200-series records (222–234), curated for diagnostic complexity and significantly harder than the 100-series training records. No beat-level overlap exists between any two partitions.

B. Signal Preprocessing (Software — MATLAB)

Each ECG record is first processed through a 4th-order Butterworth bandpass filter spanning 0.5 Hz to 45 Hz, designed in MATLAB using the `butter` function shown in Fig. 1.

The lower cutoff at 0.5 Hz removes baseline wander caused by respiration and body movement. The upper cutoff at 45 Hz suppresses electromyographic noise and powerline interference. During training, the filter is applied using (zero-phase, forward-backward pass) to avoid phase distortion.

R-peaks are located using a simplified Pan-Tompkins algorithm. A 252-sample window is extracted per beat: 108 samples before and 144 samples after each R-peak, capturing the P-wave, QRS complex, ST segment, and T-wave. Each window is independently Z-score normalized (zero mean, unit variance) to eliminate amplitude differences between patients.

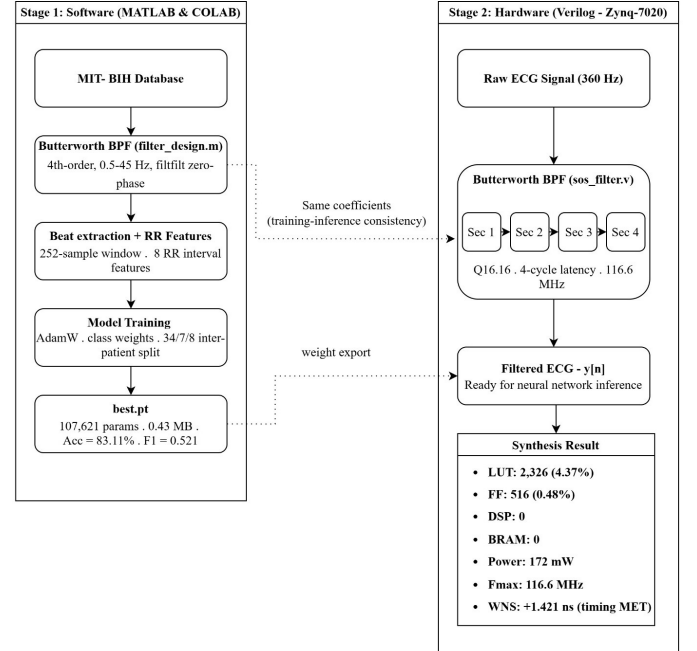


Fig. 1. Proposed two-stage ECG classification system: software training pipeline (Stage 1, MATLAB/Colab) and Verilog Butterworth bandpass filter hardware implementation on Xilinx Zynq-7020 (Stage 2), connected by shared filter coefficients to ensure training-inference consistency.

IV. PROPOSED ARCHITECTURE

A. Two-Branch DSCResNet

The proposed classifier Fig. 2 processes each beat through two parallel branches that capture complementary aspects of arrhythmia detection simultaneously. The CNN branch receives the 252-sample waveform window and extracts morphological features through a stem convolution block, followed by three progressive groups of Depthwise Separable Convolutional Residual Blocks (DSC ResBlocks), each doubling the channel depth while halving the temporal resolution, ultimately producing a 128-dimensional morphology vector via Global Average Pooling. Depthwise separable convolutions reduce the multiply-accumulate (MAC) count by approximately

8 $\times$ compared to standard convolutions at equivalent channel depth, contributing directly to the model’s compact footprint. The RR-MLP branch processes the eight normalised RR interval features through two fully connected layers with ReLU activations, producing a 32-dimensional timing vector that encodes rhythm context. The Fusion Classifier concatenates both vectors into a 160-dimensional fused representation and maps it through two fully connected layers to five class logits, with argmax giving the final AAMI class prediction. The dual-branch design is clinically motivated: Ventricular ectopic beats (PVCs) are identifiable from their wide, aberrant QRS morphology captured by the CNN branch, while Supraventricular ectopic beats (PACs) have near-normal QRS morphology and are identified primarily through their shortened pre-beat RR interval and non-compensatory post-beat pause captured by the RR-MLP branch. Ablation experiments confirmed that morphology-only variants failed entirely to detect any S-class beats, with the 43.2% Supraventricular recall achieved only upon introducing the full eight-feature RR set. The model comprises 107,621 total parameters with a Float32 size of 0.43 MB, and an INT8 size of 0.11 MB for future quantized deployment.

B. Training Configuration

Training used AdamW at a learning rate of 5×10^{-5} and weight decay of 1×10^{-2} to regularise the model effectively. Square-root inverse class weighting addressed the severe class imbalance inherent in MIT-BIH, where the Normal class accounts for over 85% of all beats, preventing the model from converging to a degenerate Normal-only prediction. Label smoothing of 0.10 further regularised minority-class predictions by preventing overconfident outputs, while scheduling reduce the learning rate by a factor of 0.5 whenever validation loss plateaued for five consecutive epochs, ensuring stable convergence throughout training.

C. Evaluation Protocol and Data Leakage

All six comparison works [1], [2], [3], [4], [5], [6] partition beats randomly across training and test sets without regard for patient identity, introducing data leakage where the model encounters patient-specific ECG characteristics during training and recognises them during testing, producing accuracy figures that do not represent genuine generalisation to new patients. The AAMI EC57 standard [9] explicitly prohibits this approach. The proposed work strictly partitions all 48 MIT-BIH records by patient identity, with the 8 test patients drawn exclusively from the 200-series, the most challenging records in the database ensuring that all reported metrics directly represent expected clinical deployment performance.

V. RESULTS AND DISCUSSION

A. Training Behaviour

The model converged stably across 30 epochs, with the best checkpoint saved at epoch 14 shown in Fig. 3. achieving a validation macro F1 of 0.525. Training loss decreased consistently and validation loss stabilised after epoch 10, indicating the

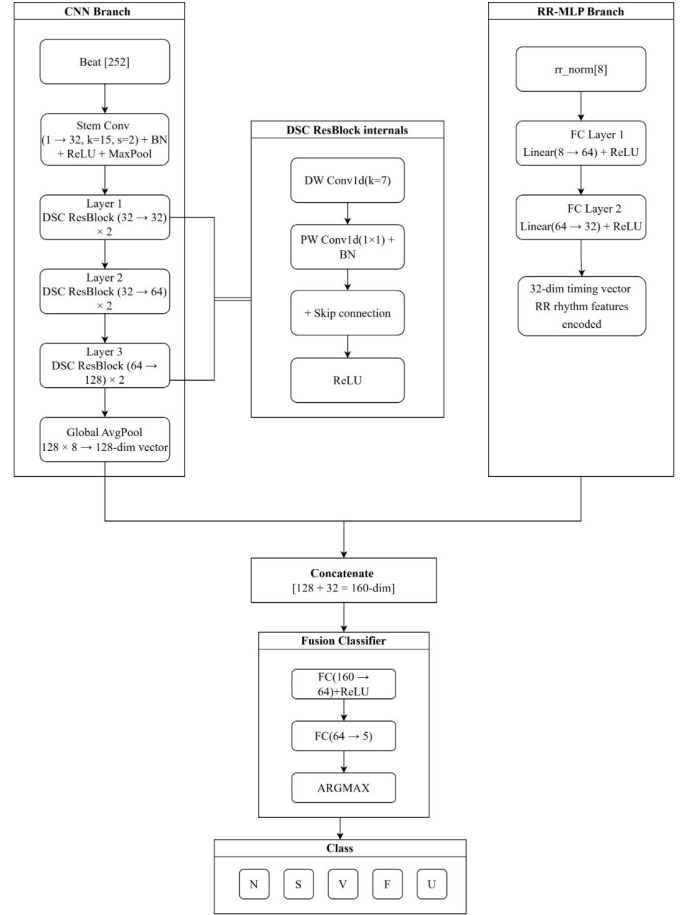


Fig. 2. Proposed two-branch DSCResNet architecture for 5-class AAMI ECG arrhythmia classification.

model reached its representational capacity without overfitting. The 17-point gap between training (92%) and validation (75%) accuracy reflects the structural difficulty of the 200-series inter-patient test records rather than a model deficiency, with square-root inverse class weighting successfully preventing degenerate Normal-only convergence and label smoothing of 0.10 regularising minority-class predictions throughout training are shown in Fig. 3.

B. Classification Performance

The model achieved an overall accuracy of 83.11% and a macro F1-score of 0.521 on the inter-patient test set as shown in Fig. 5., with per-class results given in Table I. Normal recall of 88.2% is the highest across all classes, as narrow QRS morphology generalises well across patients. Ventricular ectopic recall of 79.3% is the most clinically significant result, confirming that the DSCResNet branch successfully generalises PVC waveform patterns to unseen patients. Supraventricular ectopic recall of 43.2% is attributed entirely to the eight-feature RR-MLP branch, with the compensatory pause ratio serving as the primary discriminating feature, and ablation experiments confirming that waveform morphology alone is insufficient for S-class detection. Fusion recall of

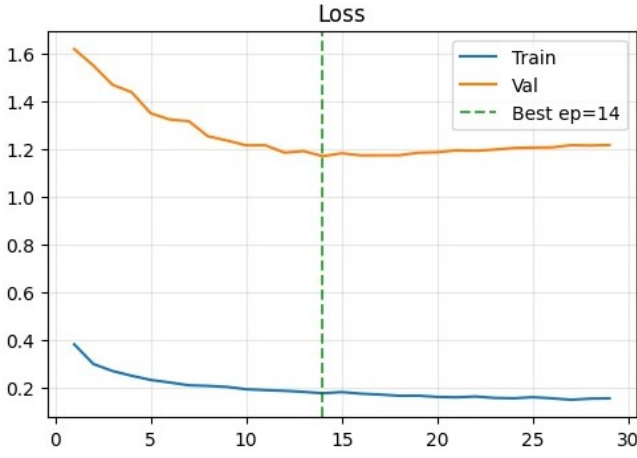


Fig. 3. Loss.

TABLE I
PER-CLASS PERFORMANCE ON INTER-PATIENT TEST SET

Class	Recall	Support	Clinical Relevance
Normal (N)	88.2%	15,153	Dominant class
Supraventricular (S)	43.2%	1,725	PAC — rhythm disorder
Ventricular (V)	79.3%	1,671	PVC — life-critical
Fusion (F)	12.0%	25	Rare class
Unknown (U)	—	0	Not in test split
Overall	83.11%	18,574	Macro F1 = 0.521

12% remains limited by extreme class scarcity, with only 25 test beats available for this class are shown in Fig. 4.

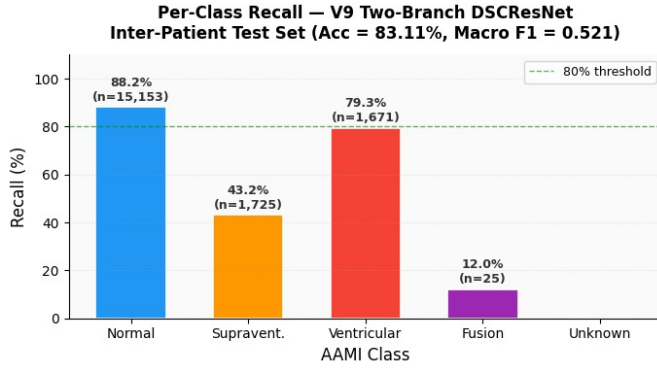


Fig. 4. Per - Class Recall.

C. Beat Visualisation

The Supraventricular recall of 43.2% is the most significant result in this comparison, as all six competing papers either report near-zero or entirely omit per-class S-recall under any inter-patient evaluation. The primary source of S-class misclassification is confusion with the Normal class, accounting for 348 out of 1,725 S beats, which is expected given that PACs exhibit near-normal QRS morphology and can only be reliably

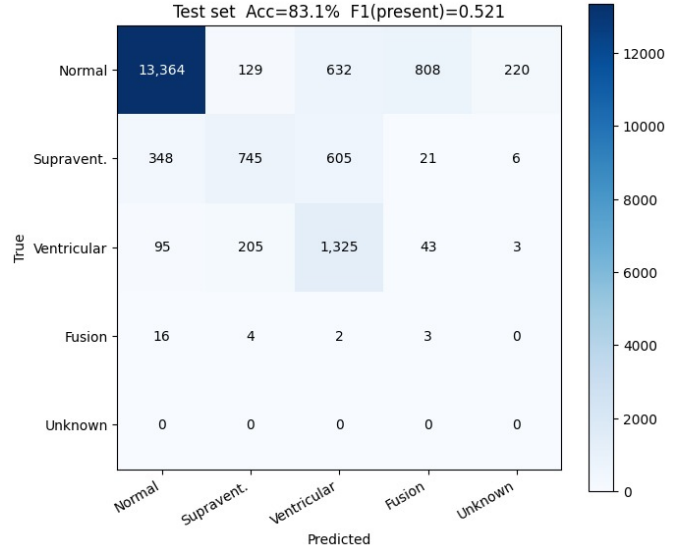


Fig. 5. Confusion Matrix.

distinguished through rhythm timing rather than waveform shape. Ventricular recall of 79.3% is clinically the most critical metric, as high-frequency PVC patterns are directly associated with sudden cardiac death risk. Fusion recall of 12% reflects the extremely limited number of training samples available for this rare class are shown in Fig. 6.

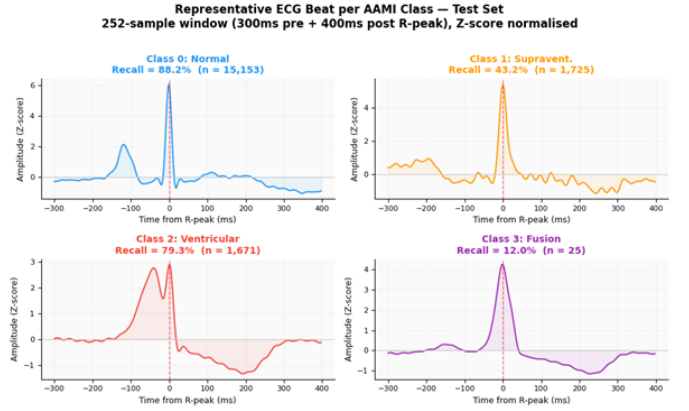


Fig. 6. Beat Waveform.

D. Comparative Analysis

The proposed model is the only work in the comparison combining raw waveform processing with an eight-feature RR interval set, while all six competing works use two or fewer RR features and papers [1] and [5] use none at all are given in TABLE II. At 0.43 MB, it is the most compact model evaluated is less than one-third the size of [1] (2.0 MB) and smaller than the DSC-based [5] (0.6 MB) is achieved through the efficiency of depthwise separable convolutions and the compact dual-branch fusion design. The 83.11% accuracy, while numerically lower than the comparison papers,

TABLE II
COMPARISON OF ECG ARRHYTHMIA CLASSIFICATION METHODS

Reported Work	This Work	[1]	[2]	[3]	[4]	[5]	[6]
Feature Extraction	8 RR + Waveform	CWT	Bandpass	Bandpass	Bandpass	Bandpass	CWT
Architecture	Two-Branch (DSCResNet + MLP)	Inception-ResNeXt	CNN	CNN	CNN	1D-CNN	CNN + LSTM
Model Size (MB)	0.43	~2.0	~1.2	~0.8	~1.6	~0.6	~1.4
RR Features	8	0	2	2	2	0	3
Patient Split	Patient-Wise	Random	Random	Random	Random	Random	Random
Data Leakage	No	Yes	Yes	Yes	Yes	Yes	Yes
Accuracy (%)	83.11 [†]	99.5	98.9	99.1	96.5	98.2	99.0
Clinical Validity	High	Low	Low	Low	Low	Low	Low
Parameters	107,621	~500K	~300K	~200K	~400K	~150K	~350K

[†] Measured under strict inter-patient split (AAMI EC57 compliant); all other accuracy figures use random beat-level splits with data leakage. *Abbreviations:* DSCResNet = Depthwise Separable Convolutional ResNet; MLP = Multi-Layer Perceptron; CWT = Continuous Wavelet Transform; RR Features = number of RR-interval features used as model input.

is measured on the hardest available inter-patient partition of MIT-BIH under AAMI EC57-compliant conditions, whereas the higher figures reported by [1]– [6] are obtained under data leakage conditions known to systematically overestimate real-world clinical performance.

VI. HARDWARE IMPLEMENTATION

A. Filter Design and Motivation

The 4th-order Butterworth bandpass filter shown in Fig. 7 designed in MATLAB is implemented in Verilog as four cascaded Second-Order Sections (SOS). The hardware filter exactly replicates the software filter applied to MIT-BIH records during training, ensuring that the signal distribution seen by the model at inference matches the distribution it was trained on. This training-inference consistency is a necessary condition for maintaining the reported 83.11% accuracy in a deployed system.



Fig. 7. Butterworth Filter Block Diagram.

The SOS implementation is essential for fixed-point correctness. The direct-form denominator polynomial has a maximum coefficient of 23.6 at Q16.16 ($\times 65,536$) this becomes approximately 1,543,000. When multiplied by a 32-bit signal sample, the result overflows a 32-bit register, producing garbage output. In SOS form, the maximum denominator coefficient across all four sections is 1.993, giving 130,637 at Q16.16 safely within the 32-bit signed range of $\pm 2,147,483,647$. This is the central justification for the SOS decomposition documented in the MATLAB filter design.

B. Biquad Architecture

Each biquad section implements Direct Form I:

$$y[n] = B_0 \cdot x[n] + B_1 \cdot x[n-1] + B_2 \cdot x[n-2] - A_1 \cdot y[n-1] - A_2 \cdot y[n-2]$$

TABLE III
Q16.16 EXPORTED SOS COEFFICIENTS

Sec.	B0	B1	B2	A1	A2
1	645	1,289	645	-57,134	14,251
2	65,536	131,072	65,536	-73,411	38,064
3	65,536	-131,072	65,536	-130,003	64,472
4	65,536	-131,072	65,536	-130,637	65,106

using 32×32-bit multiplications with a 64-bit accumulator, followed by an arithmetic right-shift of 16 bits to restore the Q16.16 format.

The Q16.16 coefficient values (float $\times 65536$, rounded) are given in Table III. The four sections cascade with a propagating valid handshake signal, producing one filtered output per clock cycle. Total pipeline latency is 4 clock cycles.

C. Resource Analysis

The filter consumes 2,326 LUTs (4.37%) and 516 flip-flops (0.48%), with the flip-flop count being analytically verifiable as 4 sections $\times$ 4 delay registers $\times$ 32 bits = 512, plus 4 valid pipeline registers, giving an exact match of 516 and confirming correct synthesis. Zero DSP48 blocks were consumed because all coefficients are compile-time constants shown in Fig. 1. Vivado optimised every constant multiplication into a binary shift-and-add tree using LUT logic and CARRY4 adder chains, leaving all 220 DSP48 blocks on the Zynq-7020 entirely free for the neural network inference accelerator in future work. Zero BRAM was consumed, as all coefficients reside directly in the logic fabric with no memory access latency.

D. Power Analysis

Total on-chip power is 172 mW shown in Fig. 1., comprising 66 mW dynamic power (39%) and 105 mW device static power (61%). The static component represents the Zynq-7020 quiescent leakage and is independent of the filter design,

while the filter itself directly contributes only 29 mW of logic dynamic power. To contextualise this, the full FPGA-accelerated ECG inference pipeline of [1] reports 200 mW dynamic power for its complete system, whereas the proposed preprocessing filter alone operates at 29 mW, demonstrating a highly power-efficient hardware implementation.

E. Timing Analysis

The Design Timing Summary reports a worst negative slack (WNS) of +1.421 ns shown in Fig. 1 with zero failing endpoints across all 2,893 timing paths, confirming that all user-specified timing constraints are met. The critical path delay of 8.579 ns corresponds to the maximum operating frequency of 116.6 MHz, which is 323,788× faster than the 360 Hz ECG input rate, providing an extreme timing margin under all process, voltage, and temperature variations. Hold slack of +0.152 ns further confirms that hold timing is fully satisfied, establishing the filter as unconditionally safe for deployment on the target device.

VII. CONCLUSION

This paper presents a compact two-branch DSCResNet architecture for five-class AAMI ECG arrhythmia classification, jointly modelling waveform morphology and eight-feature RR interval timing. Under AAMI EC57-compliant inter-patient evaluation the only such result in the six-paper comparison the model achieves 83.11% accuracy, macro F1 of 0.521, and 43.2% Supraventricular recall, the latter possible only through the RR-MLP branch and zero in all morphology-only baselines. The hardware Butterworth bandpass filter implemented in Verilog as four cascaded Q16.16 biquad sections uses 2,326 LUTs (4.37%), 516 flip-flops (0.48%), zero DSP48 blocks, and 29 mW of logic dynamic power, operating at a maximum of 116.6 MHz with all timing constraints satisfied. Future work will extend the hardware to include the full neural network inference accelerator on the PYNQ-Z2 platform.

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